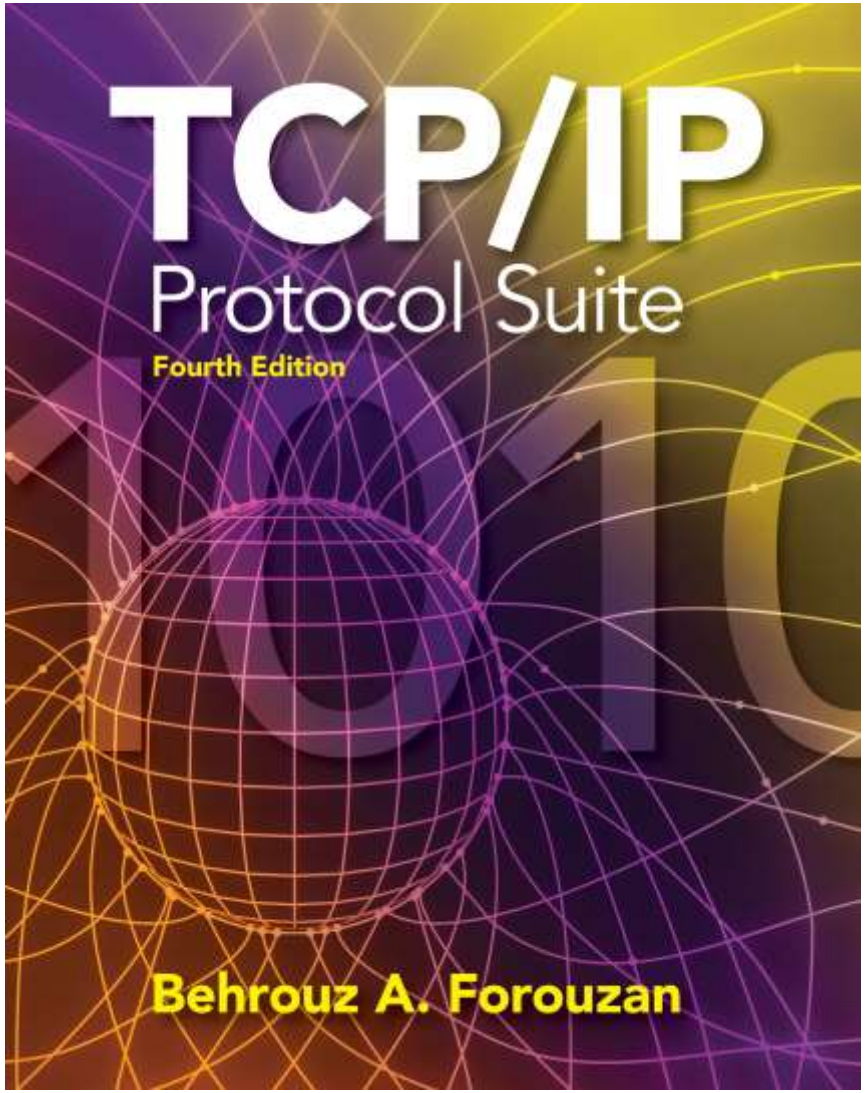
Chapter 4

Introduction to Network Layer



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OBJECTIVES:

- To introduce switching and in particular packet switching as the mechanism of data delivery in the network layer.
- To discuss two distinct types of services a packet-switch network can provide: connectionless service and connection-oriented service.
- To discuss how routers forward packets in a connectionless packet-switch network using the destination address of the packet and a routing table.
- To discuss how routers forward packets in a connection-oriented packet-switch network using the label on the packet and a routing table.
- To discuss services already provided in the network layer such as logical addressing and delivery at the source, at each router, and at the destination.
- To discuss issues or services that are not directly provided in the network layer protocol, but are sometimes provided by some auxiliary protocols or some protocols added later to the Internet.

Chapter Outline

4.1 Introduction

4.2 Switching

4.3 Packet Switching

4.4 Network Layer Services

4.5 Other Network Layer Issues

4-1 INTRODUCTION

- **At the conceptual level, we can think of the global Internet as a black box network that connects millions (if not billions) of computers in the world together.**
- **At this level, we are only concerned that a message from the application layer in one computer reaches the application layer in another computer.**

Topics Discussed in the Section

- ✓ **General Introduction**

Figure 4.1 *Internet as a block box*

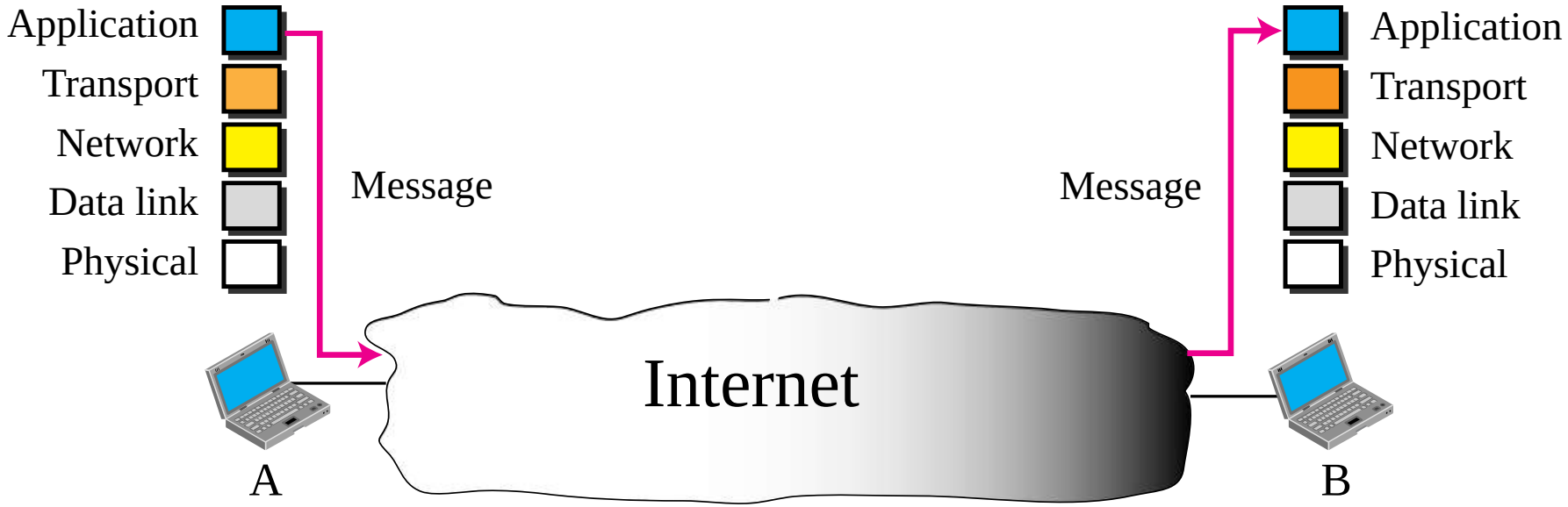
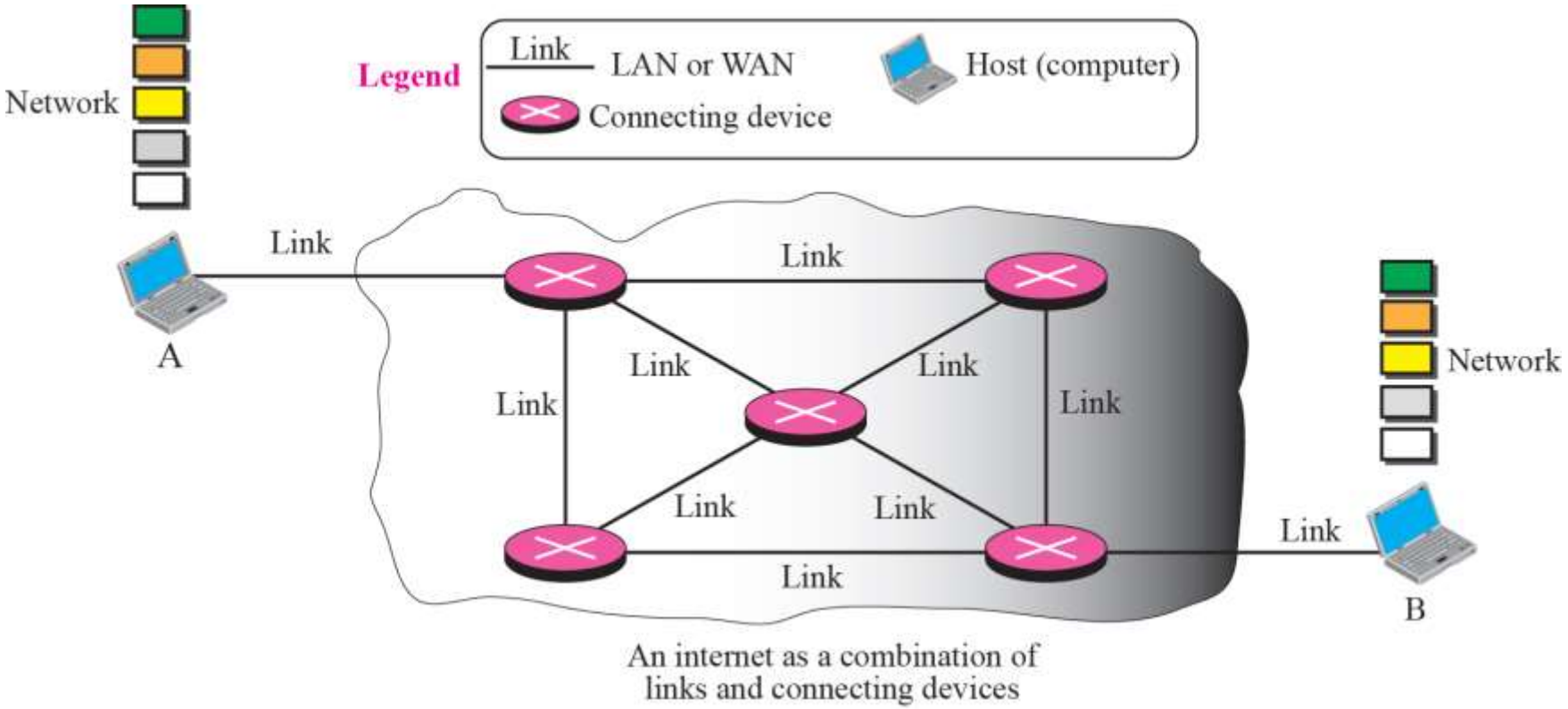


Figure 4.2 *Internet as a combination of LANs and WANs connected together*



An internet as a combination of links and connecting devices

4-2 SWITCHING

- From the previous discussion, it is clear that the passage of a message from a source to a destination involves many decisions.
- When a message reaches a connecting device, a decision needs to be made to select one of the output ports through which the packet needs to be send out.
- In other words, the connecting device acts as a switch that connects one port to another port.

Topics Discussed in the Section

- ✓ **Circuit Switching**
- ✓ **Packet Switching**

Note

In circuit switching, the whole message is sent from the source to the destination without being divided into packets.

Example 4.1

- **A good example of a circuit-switched network is the early telephone systems in which the path was established between a caller and a callee when the telephone number of the callee was dialed by the caller.**
- **When the callee responded to the call, the circuit was established. The voice message could now flow between the two parties, in both directions, while all of the connecting devices maintained the circuit.**
- **When the caller or callee hung up, the circuit was disconnected. The telephone network is not totally a circuit-switched network today.**

Note

In packet switching, the message is first divided into manageable packets at the source before being transmitted. The packets are assembled at the destination.

4-3 PACKET SWITHING

- The network layer is designed as a packet-switched network.
- This means that the packet at the source is divided into manageable packets, normally called **datagrams**.
- Individual **datagrams** are then transferred from the source to the destination.
- The **received datagrams are assembled** at the destination before recreating the original message.
- The packet-switched network layer of the Internet was originally designed as a **connectionless service**, but recently there is a tendency to change this to a connection-oriented service.

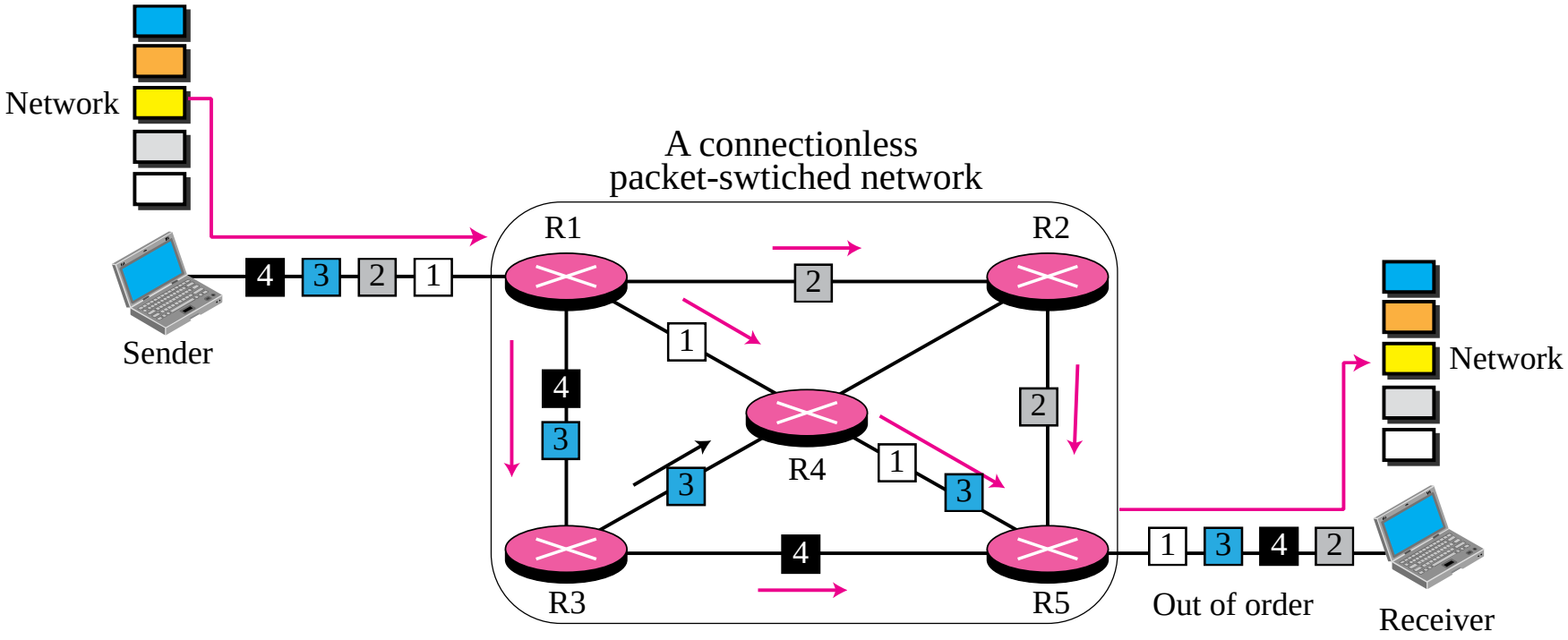
Topics Discussed in the Section

- ✓ **Connectionless Service**
- ✓ **Connection-Oriented Service**

Connectionless Service

- Network layer was designed to provide a **connectionless service,** in which the **network layer protocol** treats each **packet independently**, with each packet having no relationship to any other packet.
- The packets in a message may or may not travel the same path to their destination.
- The idea was that **the network layer is only responsible for delivery of packets from the source to the destination.**
- When the network layer provides a **connectionless service**, each **packet traveling** in the Internet is **an independent entity**; there is no relationship between packets belonging to the same message.
- The switches in this type of network are called **routers.**

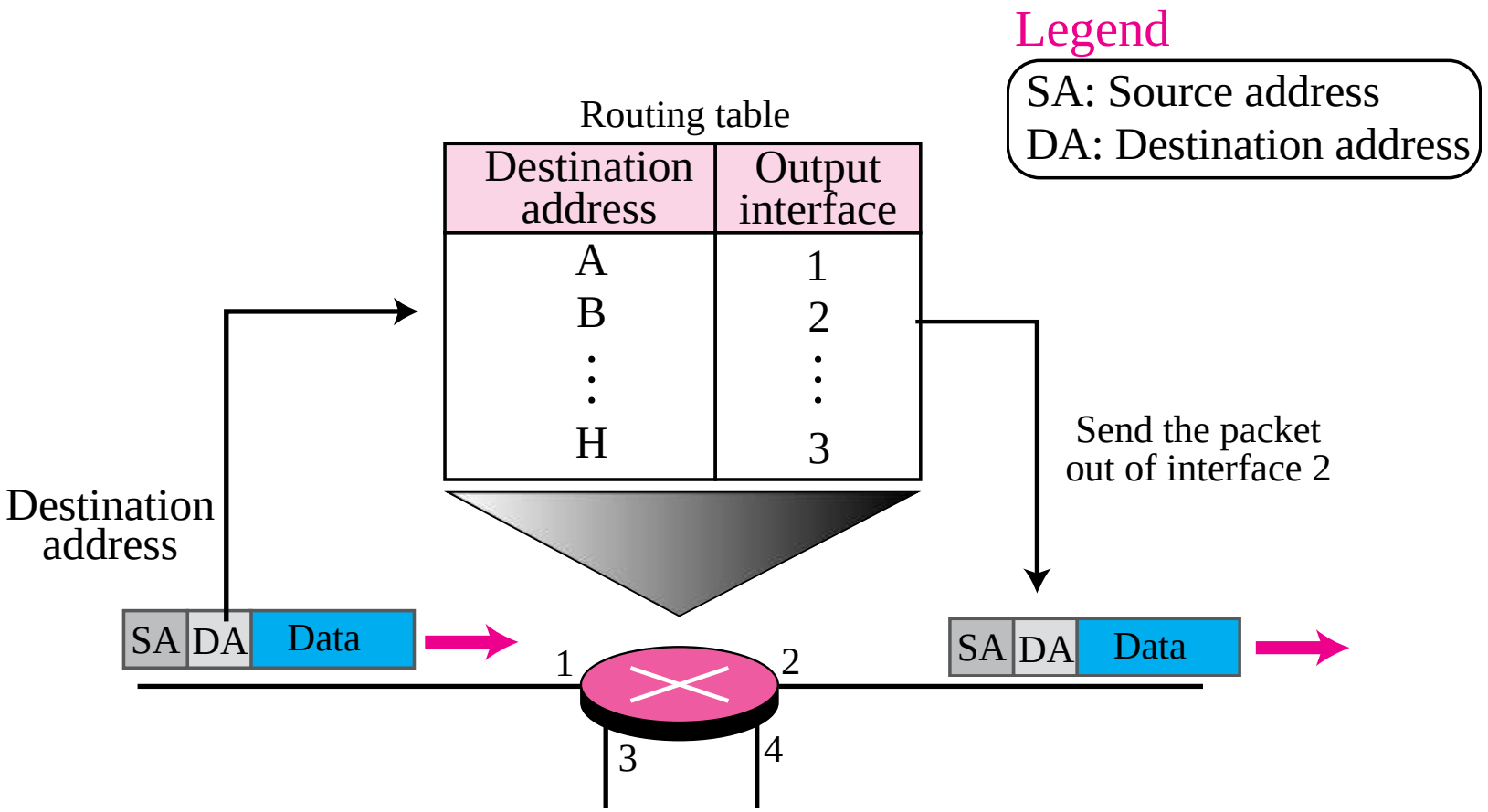
Figure 4.3 *A connectionless packet-switched network*



Forwarding Process

- Each **packet is routed based** on the information contained in its header: **source and destination address**.
- The destination address defines where it should go; the source address defines where it comes from.
- The **router** in this case routes the packet **based only on the destination address (Routing Table).**
- The source address may be used to send an error message to the source if the packet is discarded.

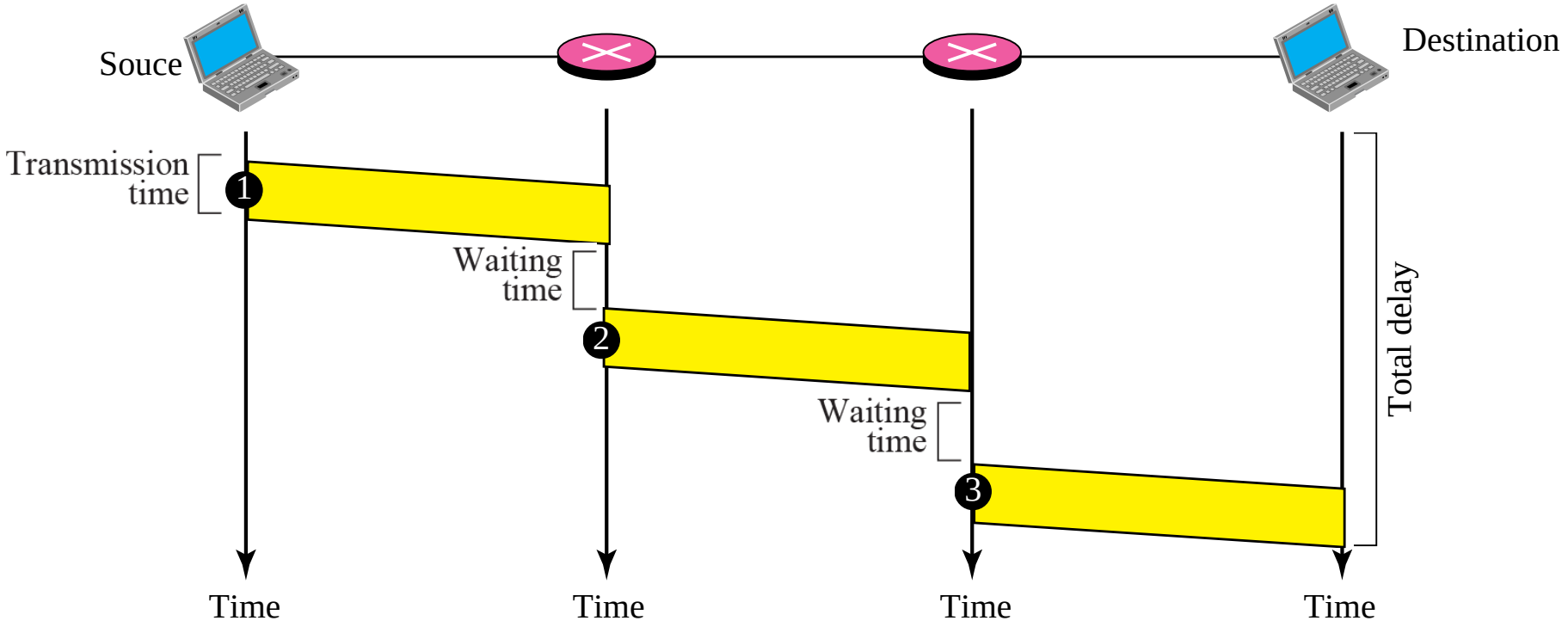
Figure 4.4 *Forwarding process in a connectionless network*



Note

In a connectionless packet-switched network, the forwarding decision is based on the destination address of the packet.

Figure 4.5 *Delay in a connectionless network*



Delay = Propagation Time + Transmission Time + Queue Time

4-4 NETWORK LAYER SERVICES

In this section, we briefly discuss services provided by the network layer. Our discussion is mostly based on the connectionless service, the dominant service in today's Internet.

Topics Discussed in the Section

- ✓ **Logical Addressing**
- ✓ **Services Provided at the Source Computer**
- ✓ **Services Provides at the Each Router**
- ✓ **Services Provided at the Destination Computer**

Figure 4.12 *An imaginary part of the Internet*

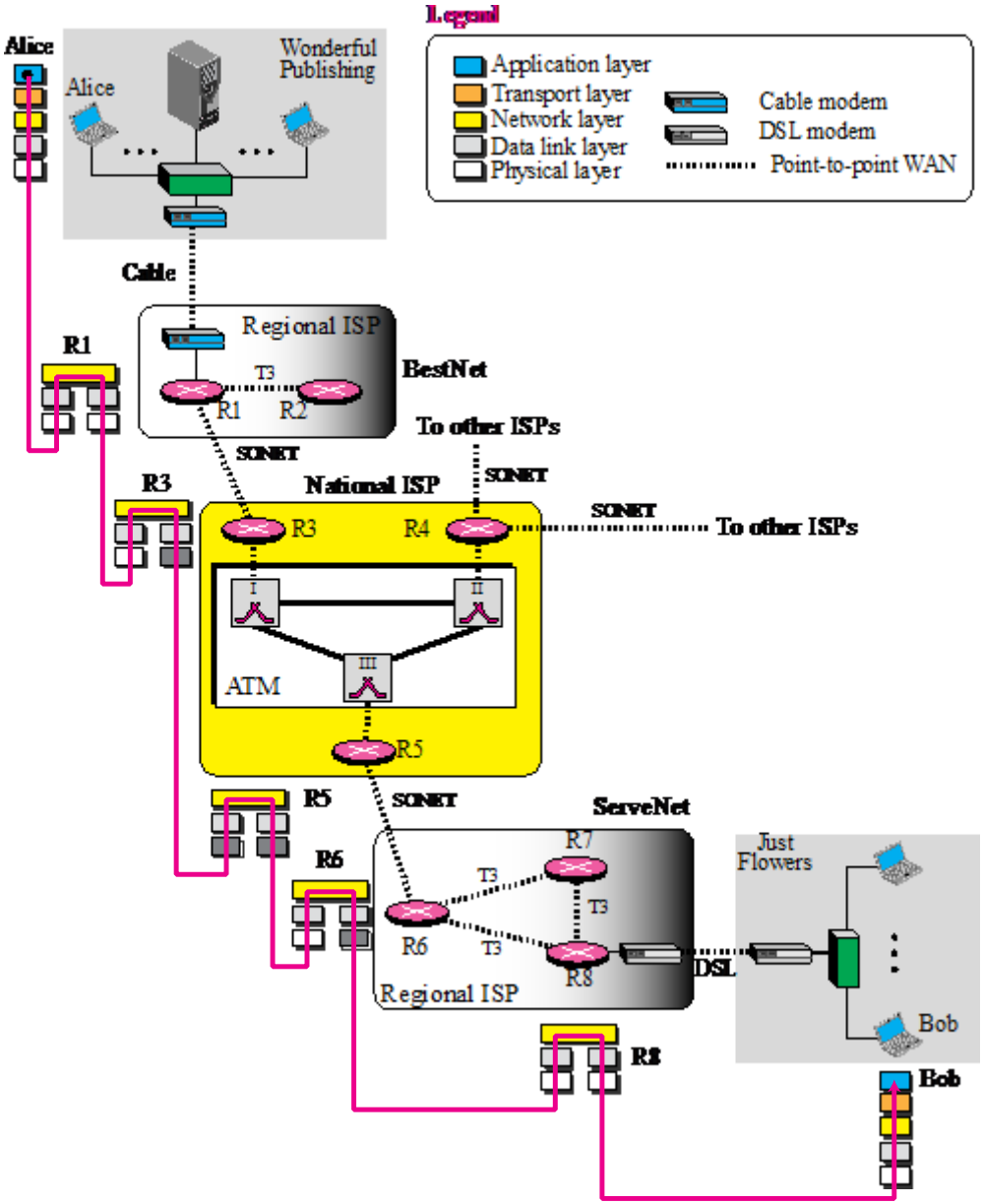


Figure 4.13 *Services provided at the source computer*

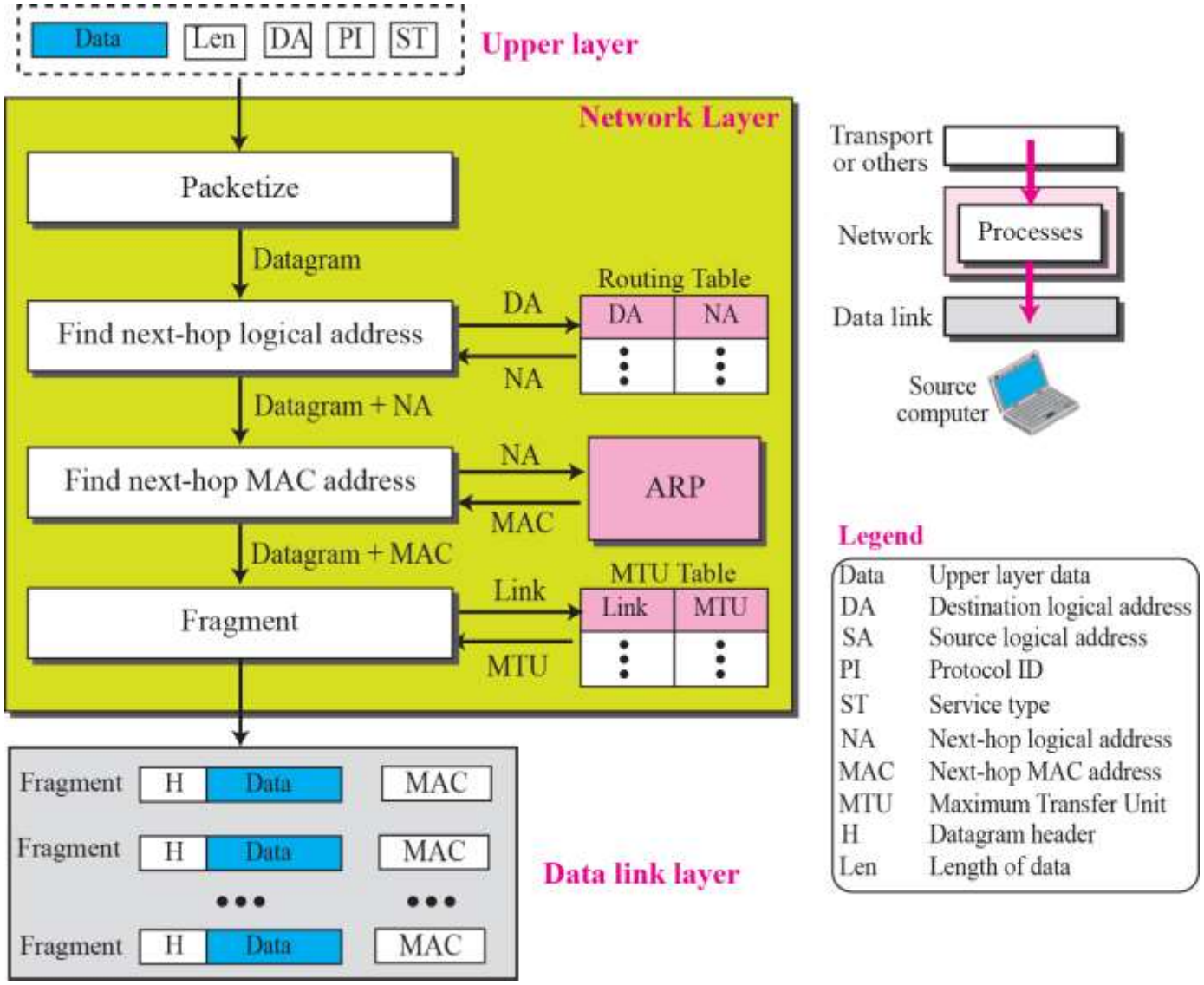


Figure 4.14 *Processing at each router*

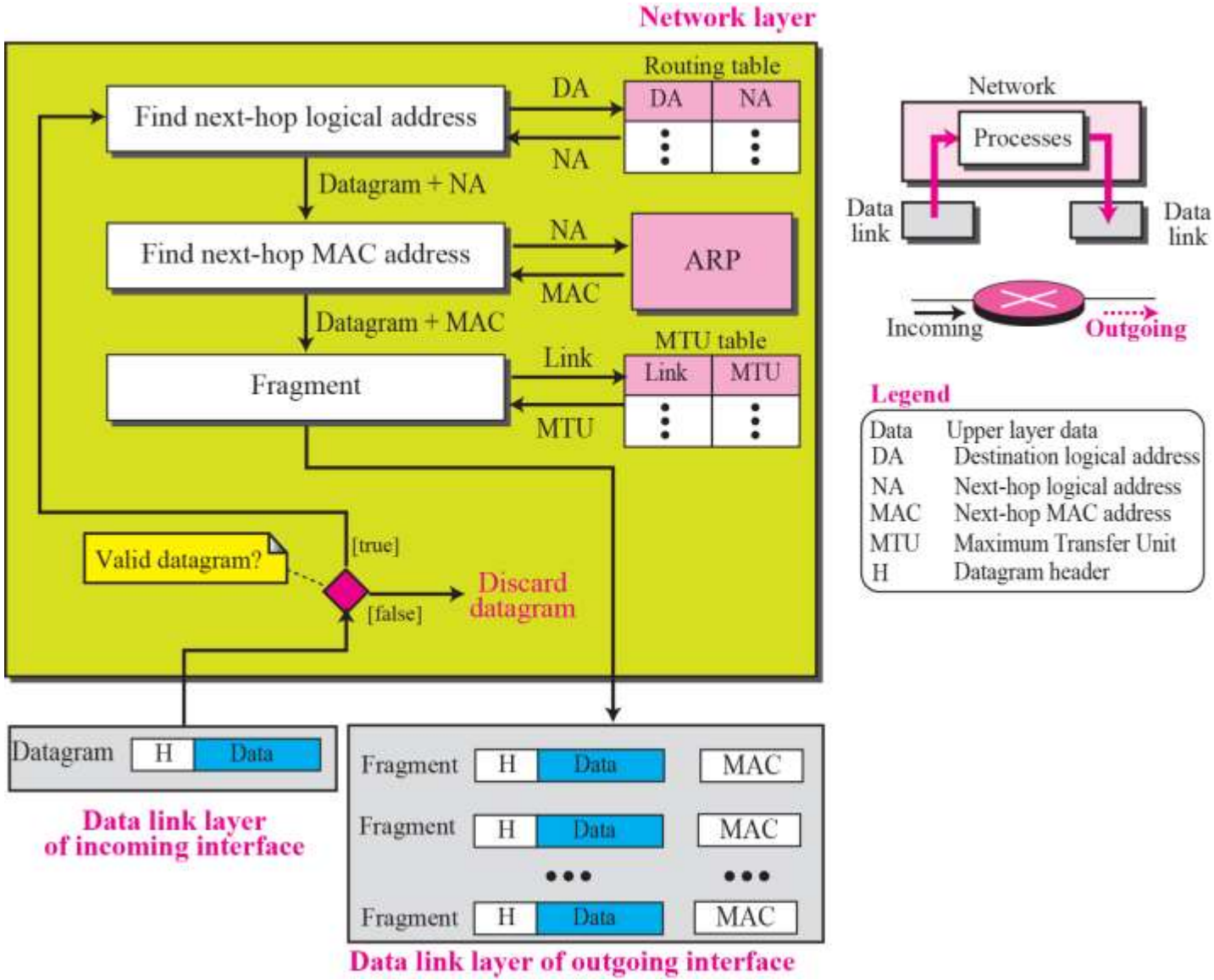
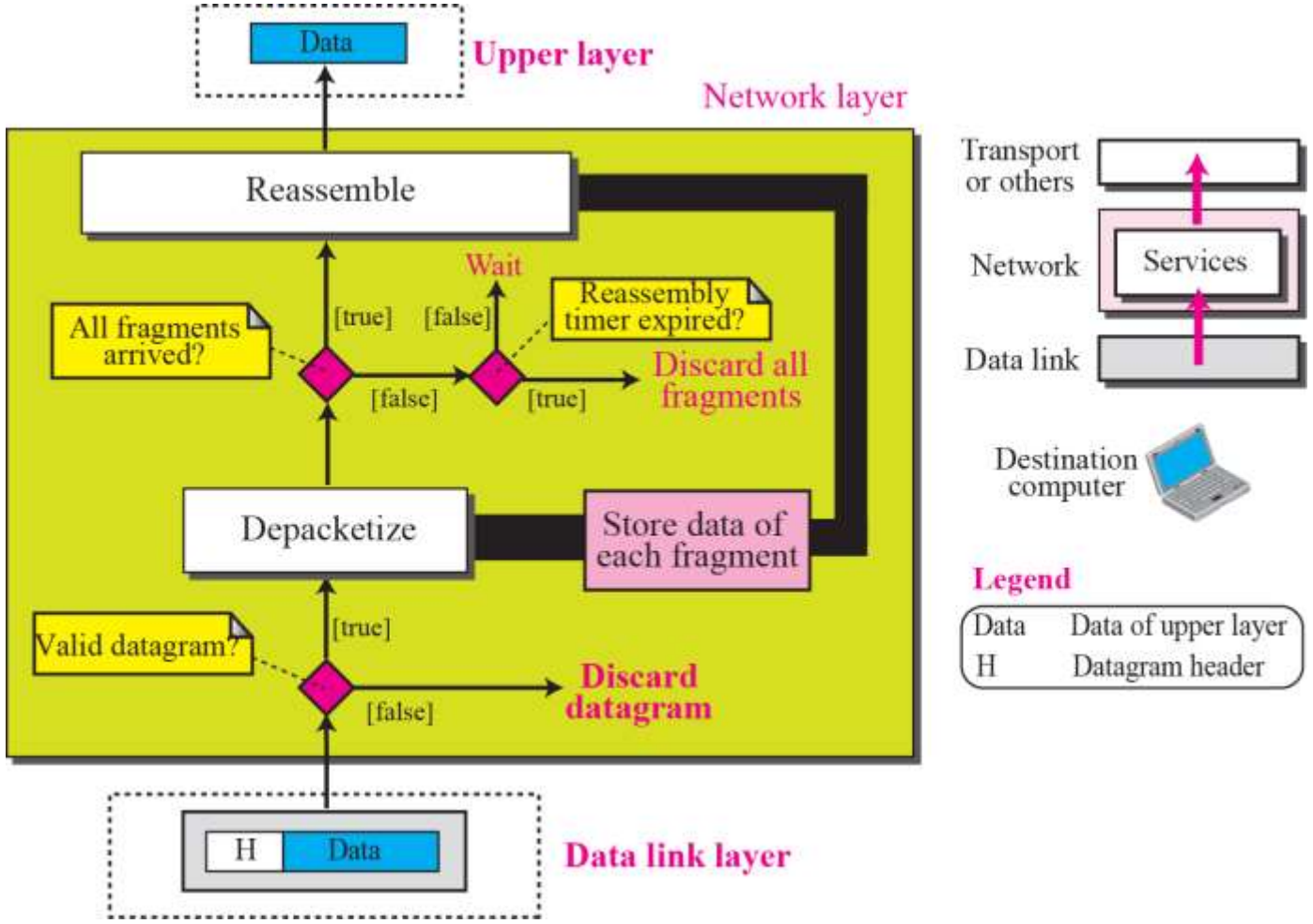


Figure 4.15 *Processing at the destination computer*



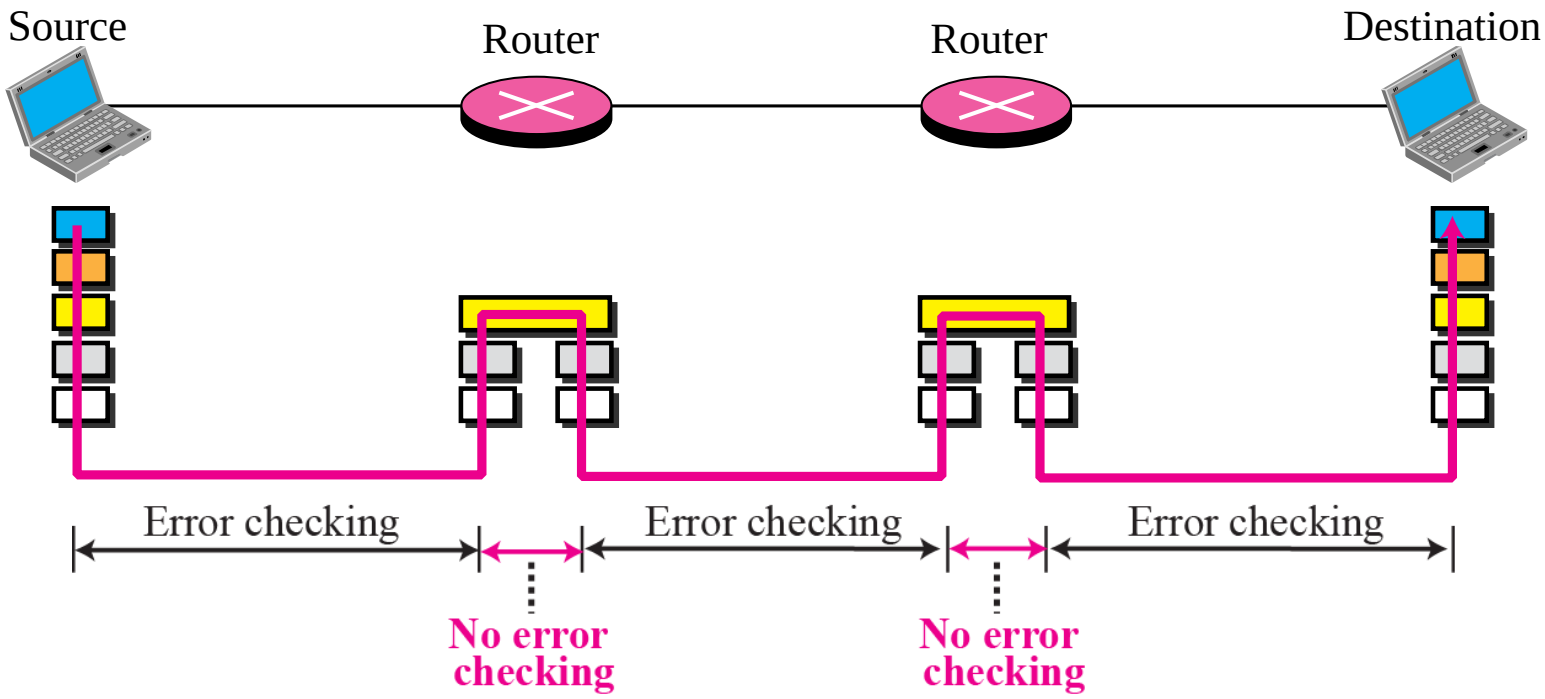
4-5 OTHER SERVICES

- In this section we introduce some issues related to the network layer.
- These issues actually represent services that are normally discussed for the network layer, but they are either partially implemented at the network layer or not implemented at all.
- Some services are provided by some auxiliary protocols or by protocols added to the Internet later.
- Most of these issues resurface in future chapters.

Topics Discussed in the Section

- ✓ **Error Control**
- ✓ **Flow Control**
- ✓ **Congestion Control**
- ✓ **Routing**
- ✓ **Security**

Figure 4.16 *Error checking at the data link layer*



Note

No flow control is provided for the current version of Internet network layer.